

AJAY BANSTOLA

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Senior Software Engineer with 5+ years of professional experience, and a strong focus on Software Engineering, Machine Learning, AI systems, and automation.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, HTML/CSS

AI/ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, CNNs, RAG, Data Analysis

GenAI Tools: LangChain, AutoGen, Hugging Face Transformers, OpenAI API, TensorBoard

Frameworks/Libraries: Django, Flask, React.js, Angular.js, Node.js, Selenium, Scrapy, BeautifulSoup

Developer Tools: Datadog, Git, Amazon Web Services (AWS), Azure Services, PyCharm, IntelliJ, VSCode, Jira, Excel

Data Visualization / BI Tools: Tableau, Power BI, Excel Dashboards

Database: SQL, MSSQL, NoSQL (MongoDB), PostgreSQL, MySQL

APIs/Integration: OpenAPI, REST APIs, SOAP APIs, Integration Testing, ETL Processes, Data Mapping and Transformation

ERP/CRM Tools: Salesforce, SAP

Others: Linux, Data Structures and Algorithms, Machine Learning, Data Science

EXPERIENCE

AI Engineer

May 2025 – August 2025

Opportunity Machine (Microsoft Azure, React js, TypeScript, ElevenLabs)

Lafayette, LA

- **AI Platform Development & Cross-functional Collaboration:** Co-led full-stack development of the **Blue Partner AI** platform, collaborating with law enforcement and engineering teams to build secure, scalable solutions. Developed a **real-time notification dispatch system** that shares live map locations with police officers, enhancing response coordination. Achieved a **75% reduction in paperwork** and enabled community-focused policing.
- **GPT Integration & Workflow Automation:** Integrated **GPT** for conversational AI, automated report generation, and policy compliance. Delivered end-user training and documentation for smooth adoption, resulting in a **70% improvement in report quality** and reducing average incident report time from **50 to 15 minutes**.

Software Engineer

November 2021 - November 2023

Smarsh Inc. (Python, Django)

Portland, OR

- Developed and deployed a scalable Web Archiving platform for **Smarsh**, automating **CI/CD pipelines** with Jenkins and reducing deployment cycles by **75%**. Maintained **95%+ unit test coverage** and integrated **Datadog** for real-time monitoring across **65+ distributed archiving servers**.
- Designed and optimized high-performance **data migration pipelines**, improving transfer efficiency by **20%** for multi-terabyte datasets from **PostgreSQL** to **MSSQL**.
- Enhanced team productivity by implementing streamlined **Git/GitHub** workflows and detailed **Confluence** documentation, increasing release coordination efficiency by **30%** across a 15+ member engineering team.

Associate Software Engineer

November 2020 – October 2021

PayNep Pvt. Ltd. (Python, Flask)

Kathmandu, Nepal

- Led the development of back-end and API services for a digital wallet, integrating services like Bus-ticketing, mobile top-up, and other payment services. Led the deployment of the **digital wallet** as well.
- Developed and documented **APIs** using Confluence, enhancing clarity and maintainability for cross-functional teams. Integrated automated test runners with **Postman collections**, improving testing efficiency and API reliability.
- Collaborated with frontend and Customer Service Representative teams to gather and refine customer requirements, ensuring seamless feature implementation.

EDUCATION

University of Louisiana at Lafayette

Lafayette, LA

Master's Of Science, Computer Science (GPA: 4.0)

January 2024- December 2025

- **Relevant Courses:** Algorithm Design, Deep Learning, Data Mining, Pattern Recognition, Programming Language, Operating System, Data Structure, Computer Architecture

Glaucoma Progression Prediction using RNFLT Maps | *Python, PyTorch, VGG, ResNet*

- Built a deep-learning pipeline to detect glaucoma progression from Retinal Nerve Fiber Layer Thickness (RNFLT) maps using the GDP500 dataset.
- Implemented and evaluated VGG and ResNet architectures as benchmark models to classify disease severity.
- Measured performance using Overall AUC, Sensitivity, Specificity, and subgroup AUCs across Asian, Black, and White patient groups.
- Improved model generalization through preprocessing, augmentation, and stratified evaluation across demographic groups.

Byteri (Binary-level Fuzzing Tool) | *C, Rust, AFL++, LLVM, Python*

- Worked under Prof. Arun Lakhotia to enhance the Byteri tool; a taint analysis and tracking framework for binary-level fuzzing.
- Developed a custom AFL++ mutator integrating Byteri’s AST-based input generation to improve code coverage and fuzzing efficiency.
- Analyzed behavioral differences between C and Rust binaries using Byteri’s taint tracking, enabling deeper insight into memory safety and data flow.

MedScan (Pneumonia detection using Image processing) | *Flask, CNN, React, Python, Machine Learning*

- Designed and worked on a web-based application to detect pneumonia using image processing techniques, implementing a Flask backend and React frontend for seamless user interaction.
- Integrated machine learning models for pneumonia detection, leveraging image processing libraries and deploying the model via Flask API for real-time predictions.

Customer Predictive Analytics | *Logistic Regression, Flask, Machine Learning*

- Developed a customer lead prediction model using logistic regression, analyzing historical customer data to identify factors influencing lead conversion. Created a website to display graphs visualizing the data results.

HONORS/AWARDS

Honors Convocation and Academic Excellence Award
University of Louisiana at Lafayette

Mar 2025

- Honored at the University’s prestigious Honors Convocation and Academic Excellence Awards for maintaining a 4.0 GPA throughout Graduate Studies (2024–2025).

Robert May Fellowship
University of Louisiana at Lafayette

Jan 2024

- Awarded the Robert May Fellowship to pursue the Master of Science in Computer Science for the full duration of the program.

CERTIFICATIONS

- Data Scientist’s Toolbox**
- Statistical Inference**
- Build a Data Science Web App with Streamlit and Python**
- Python Data Structures**
- Logistic Regression with Python and Numpy**